



MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL
(A constituent unit of MAHE, Manipal)

I SEMESTER M.TECH. (COMPUTER SCIENCE AND ENGINEERING)

END SEMESTER EXAMINATIONS, NOVEMBER - 2025

HIGH PERFORMANCE COMPUTING SYSTEMS [CSS 5103]

Date: 25/11/2025 [9:30 AM – 12:30 PM]

Time: 3 Hours

MAX. MARKS: 50

Instructions: Answer ALL the questions.

- 1A. Discuss the role of hardware and software in programmatic levels of parallel processing in parallel computer systems. Name and discuss one of the parallel processing mechanisms in which Direct memory access (DMA) operation is conducted on a cycle-stealing basis in uniprocessor system. 5M
- 1B. Explain the purpose of the *sections* directive in OpenMP. Compare it to show how it differs from the *for* directive in terms of task distribution among threads? 3M
- 1C. What is symmetric shared memory multiprocessor? Sketch its diagram and discuss its architecture. 2M
- 2A. Develop an MPI program to multiply the corresponding N elements (element-wise multiplication) for two 1-dimensional (1D) arrays. The root process reads these two array elements. The two slaves will get equally the required elements of arrays from the root that they need to handle for multiplication. Note that the root is not involved in the process of multiplication. Let each slave process print the product elements computed by them. Use only point to point communication APIs in your program. Use the user-friendly statements wherever it is needed. 5M
- 2B. Differentiate between blocking and non-blocking communication in MPI. 3M
- 2C. What are the uses of tags in MPI messages and why are they important? 2M
- 3A. There are N elements in an array to be partitioned among p number of processors in MESH SIMD where $p < N$. After partitioning, the respective processors perform the summation of the numbers allocated to them simultaneously. The availability of a parallel computer system is 2-D MESH SIMD. Construct a parallel algorithm to perform summation by the respective processors after partitioning the numbers and to obtain the final sum. 5M
- 3B. Explain in detail the following terminologies associated with SIMD computers:
 - a) Equivalence revealed between the Barrel shifter with mesh-connected network.
 - b) Routing function of inverse shuffle.5M
- 4A. There are three integer arrays A, B and C each of size 1024. Construct an OpenCL kernel to find X and Y. Here X is an array containing the elements as sum of respective elements of A and B. Y is also an array containing the elements as the sum of respective elements of X and C. In host program, assume there is already a platform id created named *platid*, a device id named *devid*, context *con* and command 5M

queue *cq*. Write the OpenCL API calls to create the buffers for X and Y. Also write the API call to invoke this kernel to meet the specified array size.

- 4B.** How will you identify the calculation of the threadId *tid* of each thread in CUDA for
- i) 2D Grid of 1D Block
 - ii) 1D Grid of 1D Block
- 4C.** Describe the role of `__kernel`, `__global`, and `__local` qualifiers in OpenCL.
- 5A.** List the steps involved in executing an OpenCL program.
- 5B.** Sketch a diagram of the CUDA device memory model for programmers for the allocation, movement, and usage of the various memory types of a device. Explain how CUDA API `cudaMemcpy()` functions with all its arguments.
- 5C.** Design an interconnection network for a SIMD computer having 16 nodes with the following routing functions where the address of a node shown is in binary. Also write the permutation cycles.

$$\beta_k (a_{n-1} \dots a_{k+1} a_k a_{k-1} \dots a_1 a_0) = (a_{n-1} \dots a_{k+1} a_0 a_{k-1} \dots a_1 a_k)$$